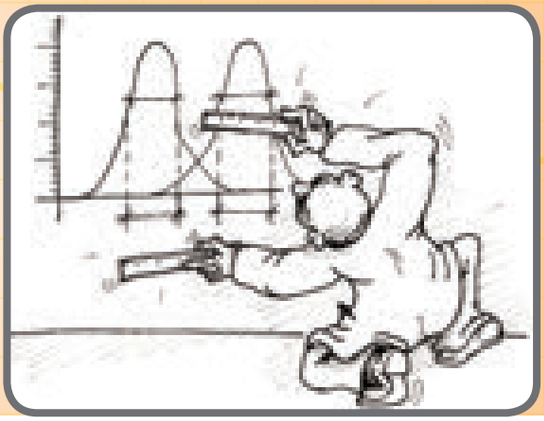
CHAPTER

3 TESTS BASED ON SAMPLING DISTRIBUTIONS – II





R. A. Fisher

Sir Ronald Aylmer Fisher (1890–1962) was a British statistician and geneticist. His work in statistics, made him popularly known as "a genius who almost single-handedly created the foundations for modern statistical science" and "the single most important figure in 20th century Statistics". In Genetics, his work used Mathematics to combine Mendelian Genetics and natural selection and this contributed to the revival of Darwinism in the early 20th century revision of the Theory of Evolution.

"Natural selection is a mechanism for generating an exceedingly high degree of improbability"

"The Best time to plan an experiment is after you have done it"

"The analysis of variance is not a mathematical theorem, but rather a convenient method of arranging the arithmetic"

LEARNING OBJECTIVES

The students will be able to

- compare variances of two populations
- understand the testing of hypothesis for comparing three or more population means.
- differentiate Treatments and Blocks.
- differentiate one-way and two-way Analysis of Variance.
- calculate *F*-ratio for Treatments and Blocks.
- infer by comparing the estimated and critical values.

AP6J38

Introduction

In the previous chapters, we have discussed various concepts used in testing of hypotheses and problems relating to means of the populations. Although many practical problems involve inferences about population means or proportions, the inference about population variances is important and needs to be studied. In this chapter we will study (i) testing equality of two population variances (ii) one-way ANOVA and (iii) two-way ANOVA, using *F*-distribution.

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